

Teo comparacion weierstrass

Teorema 1 (Criterio de comparación de Weierstrass).

Sean $(z_n)_n \subset \mathbb{C}$, $(a_n)_n \subset \mathbb{R}^+$ tales que $\forall n \in \mathbb{N} : |z_n| \leq a_n$, entonces

$$\sum_n a_n \text{ converge} \implies \sum_n z_n \text{ converge absolutamente.}$$

Demostración: Definimos $s_n := \sum_{k=0}^n z_k$, $\tilde{s}_n := \sum_{k=0}^n |z_k|$ y $A_n := \sum_{k=0}^n a_k$. Entonces, como A_n converge, es de Cauchy, por tanto $\forall \varepsilon > 0 : \exists N_0 \in \mathbb{N} : \forall n, m \geq N_0 :$

$$|\tilde{s}_m - \tilde{s}_n| = \sup_{m \geq n} \left| \sum_{k=n+1}^m |z_k| \right| = \sum_{k=n+1}^m |z_k| \leq \sum_{k=n+1}^m a_k = |A_m - A_n| < \varepsilon.$$

Por tanto, $(\tilde{s}_n)_{n \in \mathbb{N}}$ es de Cauchy y $\sum_n |z_n|$ converge. ■

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